

LESSONS FROM ONE field can often teach us something important about unrelated fields. Take the billion-year history of ice ages, and what they teach us about growing your money.

Our scientific knowledge of Earth is younger than you might think. Understanding how the world works often involves drilling deep below its surface, something we haven't been able to do until fairly recently. Isaac Newton calculated the movement of the stars hundreds of years before we understood some of the basics of our planet.

It was not until the 19th century that scientists agreed that Earth had, on multiple occasions, been covered in ice.¹⁵ There was too much evidence to argue otherwise. All over the world sat fingerprints of a previously frozen world: huge boulders strewn in random locations; rock beds scraped down to thin layers. Evidence became clear that there had not been one ice age, but five distinct ones we could measure.

The amount of energy needed to freeze the planet, melt it anew, and freeze it over yet again is staggering. What on Earth (literally) could be causing these cycles? It must be the most powerful force on our planet.

And it was. Just not in the way anyone expected.

There were plenty of theories about why ice ages occurred. To account for their enormous geological influence the theories were equally grand. The uplifting of mountain ranges, it was thought, may have shifted the Earth's winds enough to alter the climate. Others favored the idea that ice was the natural state, interrupted by massive volcanic eruptions that warmed us up.

But none of these theories could account for the cycle of ice ages. The growth of mountain ranges or some massive volcano may explain one ice age. It could not explain the cyclical repetition of five.

In the early 1900s a Serbian scientist named Milutin Milanković studied the Earth's position relative to other planets and came up with the theory of ice ages that we now know is accurate: The gravitational pull of the sun and moon gently affect the Earth's motion and tilt toward the sun. During parts of this cycle—which can last tens of thousands of years—each of the